Kirolos Sedra

RAN Data Collection Engineer / Drive Tester

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Professional Summary

RAN data collection and wireless systems engineer with hands-on experience executing **LTE/5G** field and lab measurement campaigns, including mobility testing, timestamped RF KPI collection, GPS-aligned walk tests, log validation, and post-processing. Built custom cellular data-collection workflows using **Quectel AT/QMI**, **srsRAN/Open5GS logs**, Python/Linux automation, USRP B210, and RF/network instrumentation.

Experience

Wireless Sensors and Devices Lab, University of Waterloo **Sep 2024 – Aug 2026**
Research Assistant *Full-time, On-site*

- Built and validated a private **5G SA testbed** using an **srsRAN gNB**, **Open5GS 5GC**, **USRP B210**, and **Quectel RM502Q-AE UE**; resolved AMF/SMF/UPF registration, PDU-session, and user-plane connectivity issues.
- Executed attenuation-based SNR ladders, static OTA tests, mobility scenarios, and **4G/5G walk-test data collection**; measured and logged **RSRP, RSRQ, SINR, RSSI, CQI, MCS, UL/DL throughput**, and latency under load, and aligned timestamped radio/performance logs with **GPS** data for street-level analysis.
- Built a custom cellular/RF data-collection workflow using **Quectel AT/QMI interfaces**, srsRAN gNB and Open5GS logs, **Python/Linux automation**, tcpdump/Wireshark, and post-processing/visualization to validate data quality and investigate RF, mobility, and network issues.
- Characterized indoor and outdoor **5G/LTE and Wi-Fi performance for mobile-robot teleoperation**, correlating radio-quality measurements with P99 latency and throughput to quantify mobility, interference, and LTE-anchoring effects; documented results for technical and industry stakeholders.
- Automated wireless validation for up to **30 DUTs**, reducing manual intervention by approximately **70%** through RF logging, network-service automation, SSH-based configuration, and PHY/transport analysis.

Aquasensing **Sep 2024 – Present**
Firmware Specialist / Research Intern *Part-time*

- Validated and debugged production **BLE–LTE IoT gateways** through RF-informed and anechoic-chamber testing, component stubs, and cross-layer isolation of firmware, connectivity, and cloud failures.
- Managed cellular IoT connectivity through **Hologram MVNO** and **Bell Jasper**; debugged **nRF/Zephyr** firmware and automated provisioning and validation workflows for **70 IoT gateways**.

University of Waterloo **Feb 2025 – Apr 2026**
Teaching Assistant, ECE 610 Broadband Networks and ECE 358 Computer Networks *Part-time*

Ejada Systems **Sep 2023 – Aug 2024**
Software Engineer – Java Spring Boot, REST APIs, SQL Optimization *Full-time*

Publications

CCECE 2026: Trajectory-Based Performance Characterization of Wi-Fi for Outdoor Mobile Robot Teleoperation; Scalability Analysis of Wi-Fi Broadcast Mechanisms Using Simulation and Hybrid Approaches.

Education

University of Waterloo – Master of Applied Science, Electrical and Computer Engineering Sep 2024 – Aug 2026
Grade: 92% (A+)

Ain Shams University – Bachelor of Science, Computer Engineering Sep 2018 – Jul 2023
CGPA: 3.12/4.00

Technical Skills

Cellular/RAN: 5G NR SA/NSA, LTE, RAN architecture, PHY/MAC/RRC procedures, UE registration, PDU sessions, UL/DL analysis, mobility/field testing, 3GPP TS 38-series exposure
Data Collection/Diagnostics: Quectel AT/QMI, srsRAN gNB/Open5GS logs, GPS-aligned logging, Python/Linux automation, tcpdump, Wireshark, iw, nmcli, RF data post-processing and visualization
RF/Performance: RSRP, RSRQ, SINR, RSSI, SNR, CQI, MCS, throughput, latency under load, attenuators, OTA, spectrum analyzer, TinySA, VNA, anechoic chambers, interference/coexistence
Simulation/Systems: ns-3, CTTC 5G-LENA, NR-U, GNU Radio, MATLAB, Simulink, Python, C/C++, Linux, TCP/IP, SSH